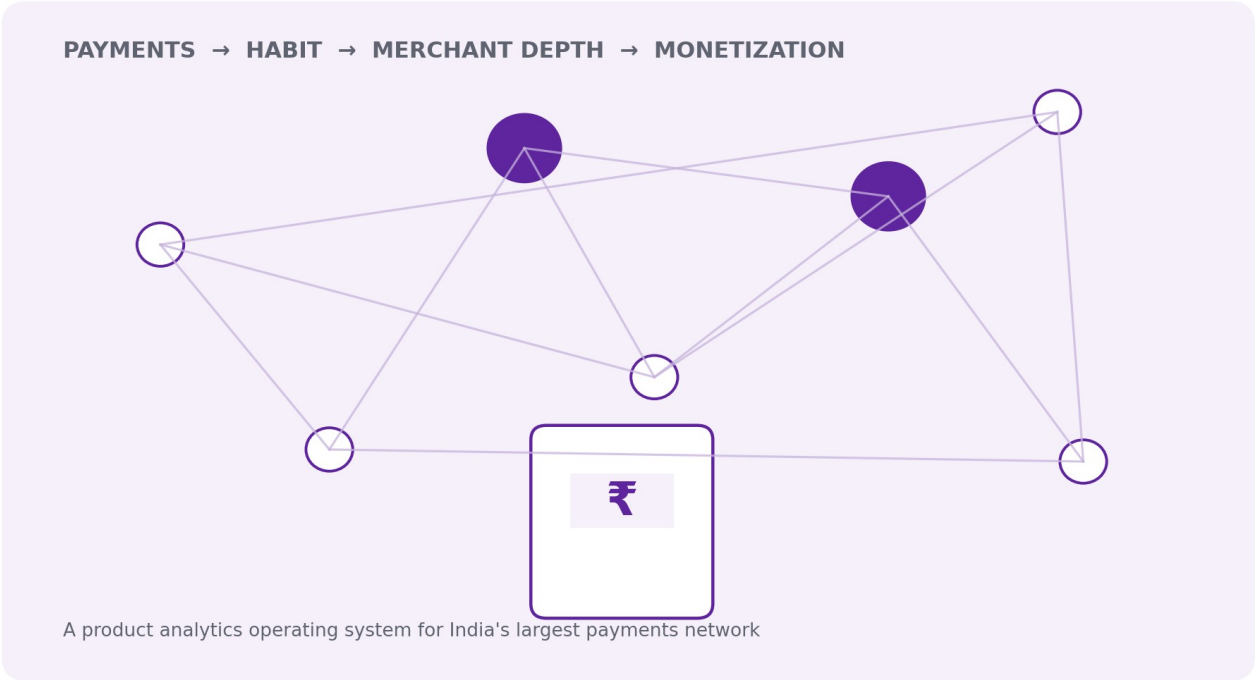


Payments Growth OS

From UPI Leadership to Active, Habitual & Monetizable Payment Relationships



711.7M	50.7M	11B	24.51B
Q2 2026 registered users	Q2 2026 registered merchants	Jul-2026 customer transactions	India UPI txns, Aug-2026

Product Analytics | Merchant Growth | Experimentation | Geographic Strategy | SQL

Research cut-off: 8 September 2026 • Public data + clearly labeled modeled analytics layer

01 Executive Summary

The strategic problem is activation quality, not registration scale.

CORE QUESTION Where should PhonePe invest next to maximize durable payments growth: registrations, activation, habit, merchant depth, monetization, or new access channels?

618.4M	230.08M	44.87M	11.31M
Registered users, Mar-25	MAC, Mar-25	Registered merchants, Mar-25	MAM, Mar-25

PhonePe already operates at population scale. Yet its filing separates lifetime registered populations from monthly active relationships. This creates a large activation-depth question - while the users and merchants who are active are simultaneously becoming more habitual.

- Consumer TPC increased from 25.50/month in Mar-2023 to 38.10 in Mar-2025; habitual users (>30 TPC) rose from 28.41% to 41.43% of MAC.
- Merchant TPAM increased from 147.87 to 379.16 between Mar-2023 and Sep-2025; DAM/MAM rose from 44.18% to 60.77%.
- Q2 2026 PhonePe Pulse shows Business payments at ~64% of transaction count but only ~23% of value - merchant payments are a high-frequency engine.
- FY25 revenue was ₹7,114.9 Cr; FY26 revenue increased to ₹7,920.5 Cr while reported net loss widened to ₹2,791.6 Cr, raising the importance of durable monetization.

Recommendation

- P0 - Merchant Launchpad: first transaction → 30-day retained merchant.
 - P0 - Geographic activation: target large, already-distributed but lower-intensity states.
 - P1 - Consumer Habit Ladder: migrate developing users into recurring multi-use-case behavior.
 - P1 - Merchant depth: devices/business tools/financial services after usage fit.
 - Strategic - UPI 123Pay: optimize feature-phone activation and retention, not pre-installs.
- Source basis: PhonePe UDRHP-I, PhonePe Pulse Q2 2026, PhonePe current releases, NPCI and current regulatory-filing coverage. Reported, derived and modeled numbers are explicitly separated.

02 Market & Current Scale

UPI is already national infrastructure; PhonePe is optimizing within a mature, enormous rail.

22.35B	23.20B	22.72B	24.51B
UPI txns Apr-26	UPI txns May-26	UPI txns Jun-26	UPI txns Aug-26

NPCI monthly statistics show UPI processing more than 22B transactions per month through Q1 FY27, and Reuters reported a new record of 24.51B transactions worth ₹29.82T in Aug-2026.

718M	50M+	11B	98%+
LTD users, Jul-26	LTD merchants, Jul-26	PhonePe customer txns, Jul-26	postal-code merchant reach

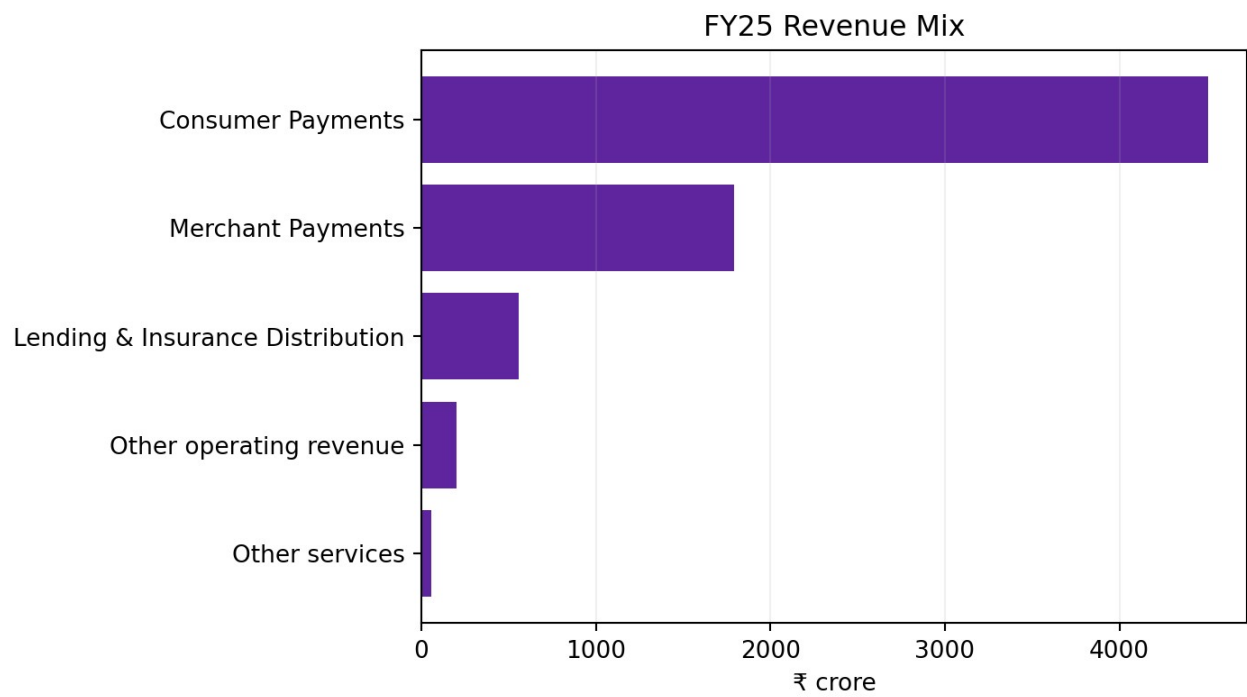
DATA DISCIPLINE PhonePe Pulse Total Transactions is not the same metric as NPCI payer-app volume. Pulse includes transactions initiated by PhonePe customers OR received by registered PhonePe merchants; market-share analysis must use NPCI/TPAP definitions separately.

Why this matters for the case

At this scale, acquisition remains useful at the margin but is no longer a sufficient growth strategy. The product system must distinguish reach, activation, retention, transaction depth, trust and revenue quality.

03 Business Model & Monetization

Payments create distribution; monetization increasingly sits on merchant and financial-service layers.



REPORTED: PhonePe UDRHP FY25 revenue streams. ₹ crore.

Stream	FY25 ₹ Cr	Share %
Consumer Payments	4,506.9	63.3%
Merchant Payments	1,792.8	25.2%
Lending & Insurance Distribution	557.6	7.8%
Other services	57.2	0.8%
Other operating revenue	200.2	2.8%

MONETIZATION SIGNAL Lending & Insurance Distribution revenue grew from ₹28.1 Cr in FY23 to ₹557.6 Cr in FY25; H1FY26 alone reached ₹452.6 Cr. Payment scale is increasingly a distribution surface for higher-monetization services.

FY26 current filing context: revenue from operations ₹7,920.5 Cr (+11.5% YoY); reported net loss ₹2,791.6 Cr. Detailed stream analysis above uses UDRHP FY25 because it provides the cleanest public breakdown.

04 KPI Architecture

Do not optimize “users” or “transactions” as isolated vanity metrics.

CONSUMER NORTH STAR Habitual Monthly Active Customers: MAC crossing the >30 transactions/month threshold used in PhonePe's filing.

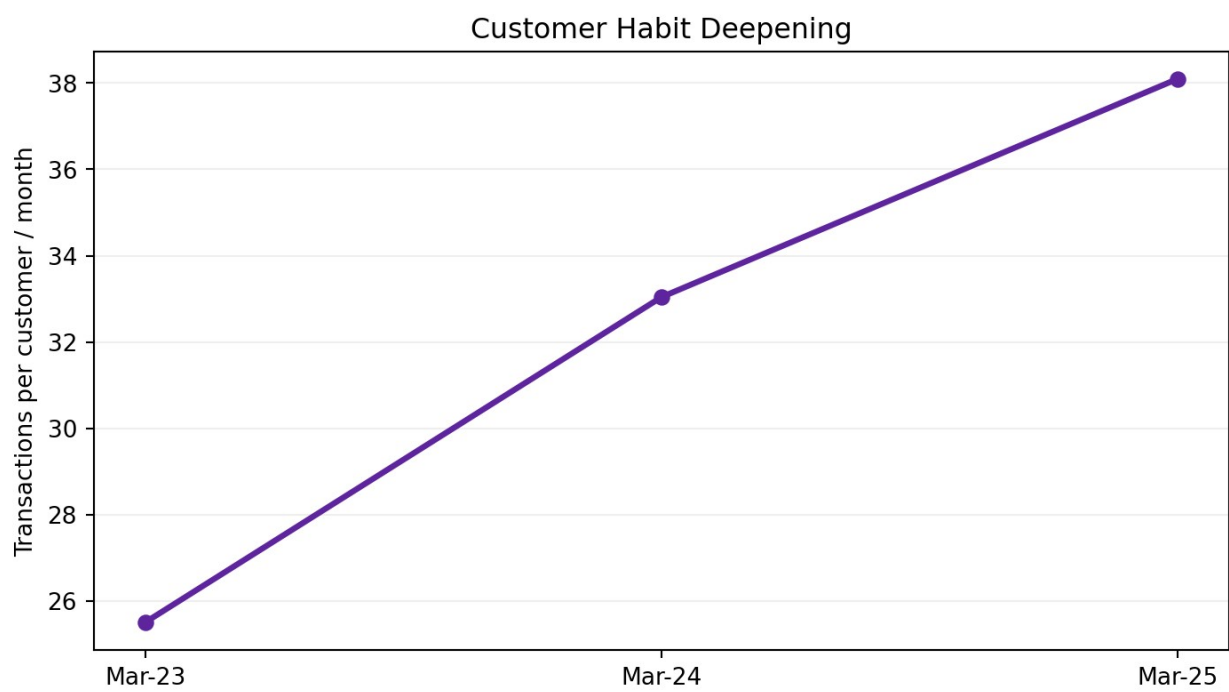
MERCHANT NORTH STAR Retained Monthly Active Merchants: transacting merchants retained across a defined cohort window, paired with TPAM to capture depth.

Layer	Metrics	Decision question
Acquisition	LTD registrations, new merchant onboardings	Reach, not quality
Activation	first successful payment, D7 active	Does the relationship become real?
Retention	M1/M3/M6 active cohort	Does usage persist?
Depth	TPC, TPAM, DAM/MAM, use-case breadth	How habitual is the relationship?
Monetization	revenue/active relationship, devices, financial services	Does activity create durable economics?
Trust	success rate, fraud, disputes, support	Can growth scale safely?

Interview principle: if a feature raises transaction count but worsens success rate, fraud, support cost or retention, it is not automatically a product win.

05 Consumer Habit Diagnosis

Active customers are transacting more frequently over time.



REPORTED: PhonePe UDRHP. TPC = monthly transactions / monthly active customer.

25.50	38.10	28.41%	41.43%
TPC Mar-23	TPC Mar-25	Habitual / MAC Mar-23	Habitual / MAC Mar-25

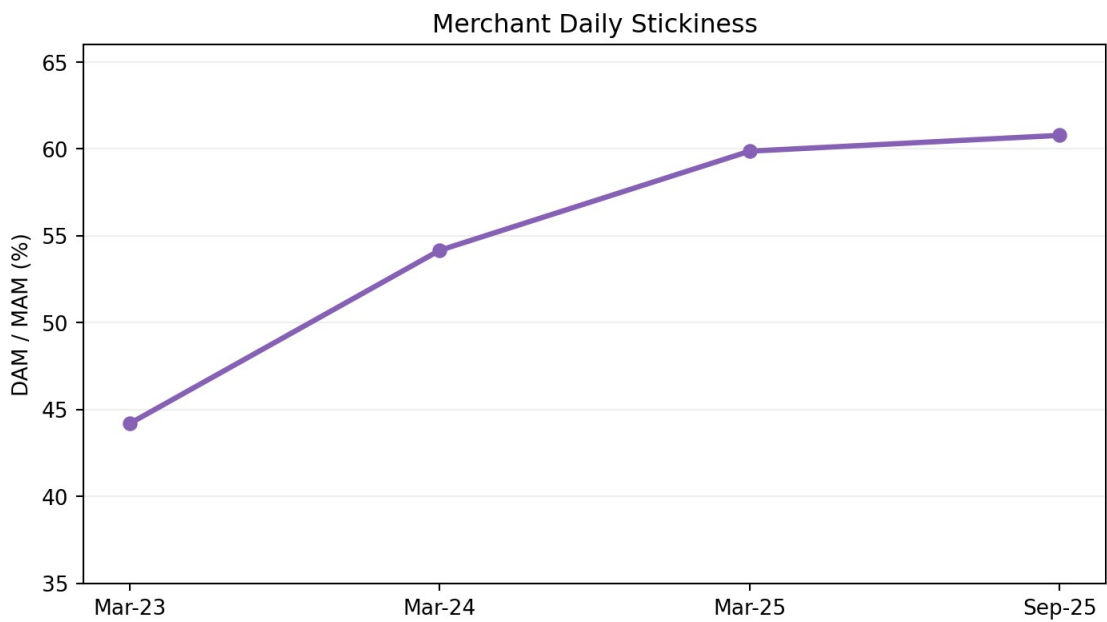
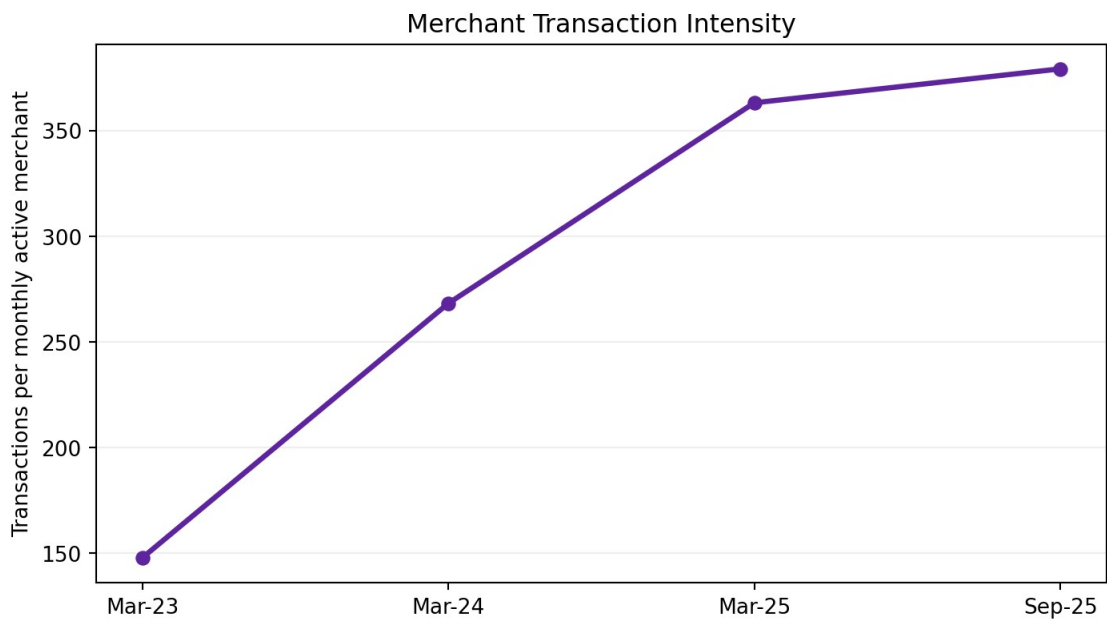
The filing states that transaction frequency increases as customer cohorts age and newer cohorts ramp faster in their first 12 months. This suggests product breadth and familiarity create habit, not merely acquisition volume.

Product implication

- Segment by current monthly frequency and dominant use case.
- For developing users, recommend one next recurring payment habit rather than the entire PhonePe surface.
- For habitual users, prioritize reliability, security and relevant financial-service discovery over generic payment nudges.

06 Merchant Engagement Diagnosis

The merchant network is becoming more daily and more transaction-dense.

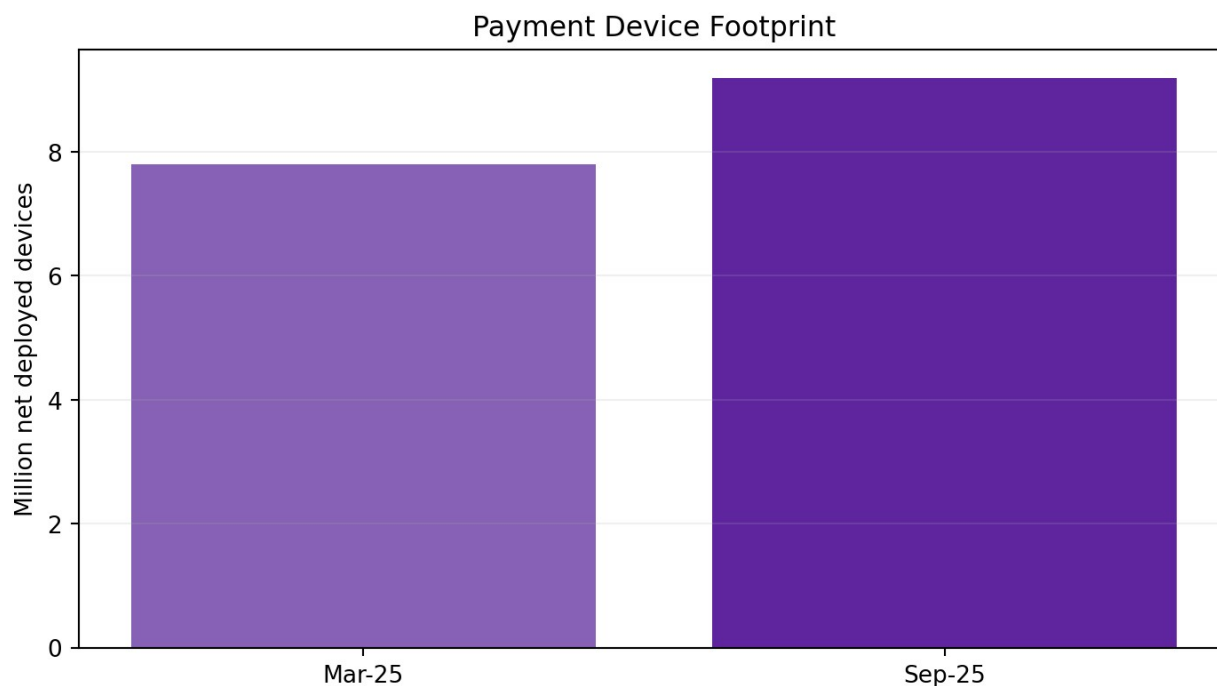


REPORTED: PhonePe UDRHP. DAM/MAM captures daily merchant stickiness; TPAM captures monthly transaction intensity.

147.87	379.16	44.18%	60.77%
TPAM Mar-23	TPAM Sep-25	DAM/MAM Mar-23	DAM/MAM Sep-25

07 Merchant Infrastructure

Physical payment devices deepen engagement and create a recurring monetization surface.



REPORTED: 9.19M net deployed payment devices as of Sep-2025.

PhonePe identifies Smartspeakers and EDC machines as a merchant-engagement strategy and a recurring-revenue source. Device deployment, however, should be treated as an eligibility decision: hardware cost is better deployed where transaction density and retention justify it.

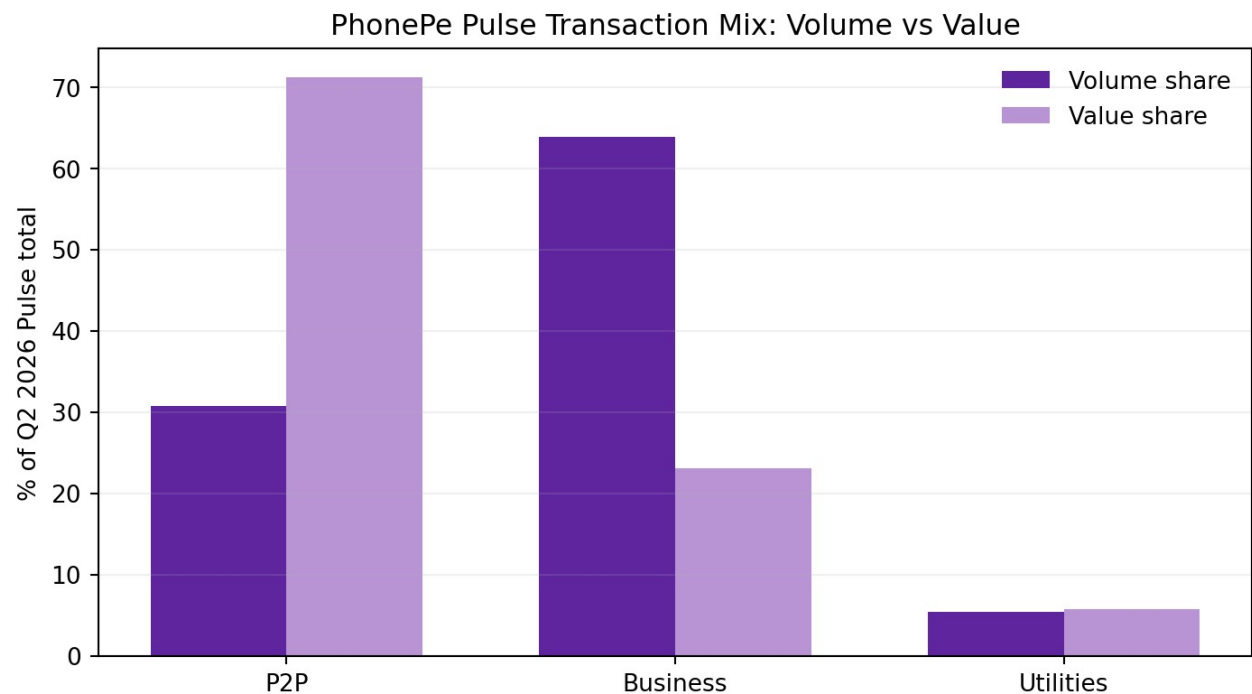
CAUSALITY WARNING If device merchants retain better, that does not prove the device caused retention. High-quality merchants may be more likely to receive a device. Use randomized eligibility, propensity matching or difference-in-differences where practical.

Eligibility signals

- TPAM and active days
- transaction stability / TPV trend
- merchant category and payment mix
- refund/dispute rate
- existing QR usage
- expected subscription/device lifetime

08 Q2 2026 Transaction Mix

Merchant/business payments dominate count; P2P dominates value.



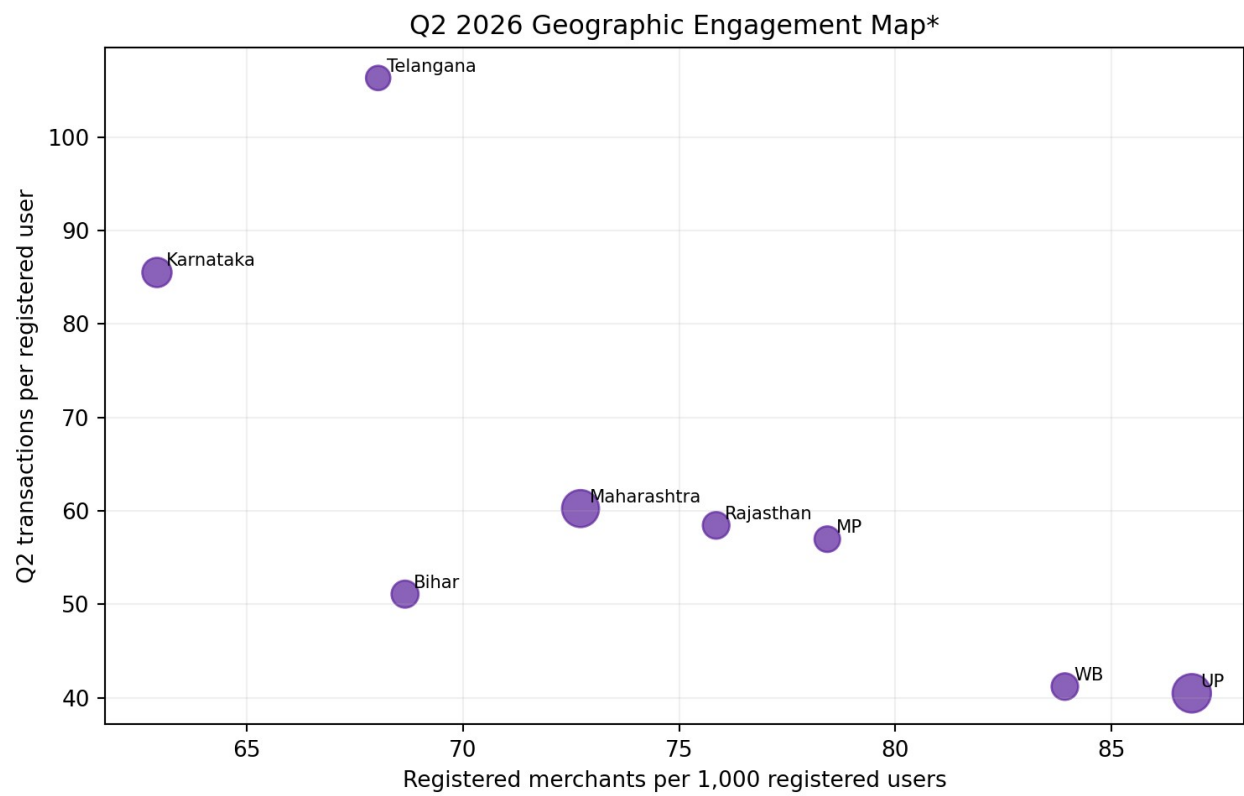
REPORTED + DERIVED from Q2 2026 PhonePe Pulse. Business/P2P/Utilities are mutually exclusive in the refreshed dataset.

Category	Tx Bn	Vol %	TPV ₹T	Value %	ATV ₹
P2P	11.89	30.8%	32.39	71.2%	2,724
Business	24.69	63.9%	10.48	23.0%	425
Utilities	2.08	5.4%	2.62	5.8%	1,259

INSIGHT Business transactions represent ~63.9% of Pulse count but ~23.0% of value, with derived ATV ~₹425. That makes merchant payments a frequency engine and a strong surface for habit, devices and merchant services.

09 Geographic Growth Lens

Distribution and engagement are not equally deep across large states.

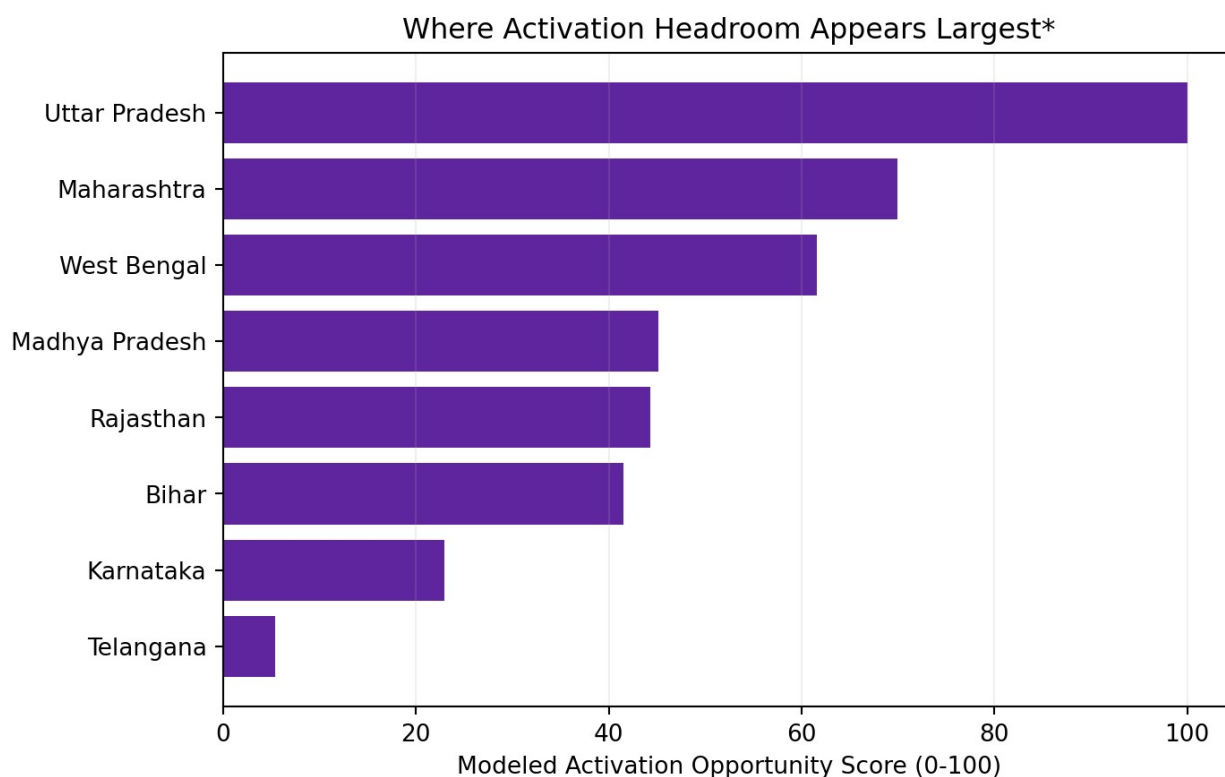


*DERIVED from Q2 2026 Pulse top-state aggregates. Ratios use LTD registered bases, not active-period denominators.

State	Tx/User Q2	Merchants/1k users	Modeled score	Read
Uttar Pradesh	40.5	86.9	100.0	Activate at Scale
Maharashtra	60.2	72.7	69.9	Grow & Deepen
West Bengal	41.2	83.9	61.6	Merchant Engagement
Madhya Pradesh	57.0	78.4	45.1	Grow & Deepen
Rajasthan	58.4	75.9	44.3	Grow & Deepen
Bihar	51.1	68.7	41.5	Grow & Deepen
Karnataka	85.5	62.9	22.9	Defend & Monetize
Telangana	106.3	68.0	5.3	Defend & Monetize

10 Activation Opportunity Index

A decision framework, not a company metric.



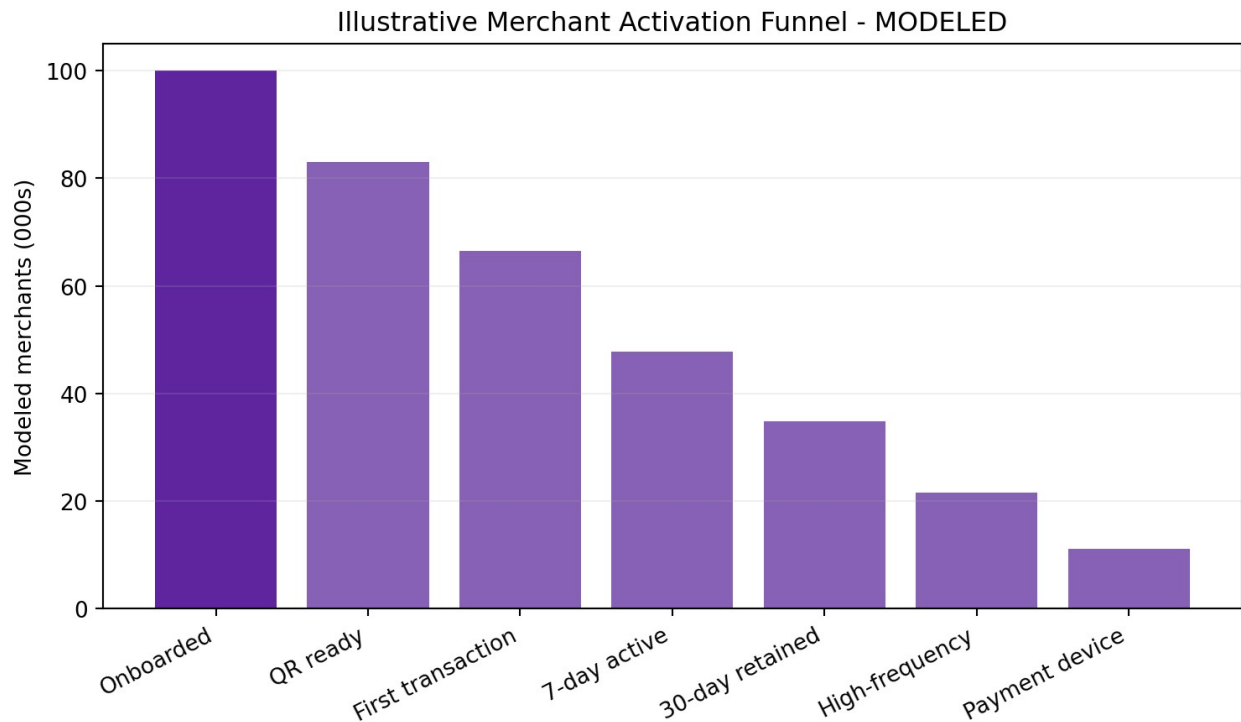
MODELED score = 40% user scale + 35% inverse transaction intensity + 25% merchant density, normalized within the displayed top-state sample.

- Uttar Pradesh: very large registered user and merchant base with comparatively low Q2 transaction intensity → activation-at-scale.
- West Bengal: high registered merchant density but lower activity intensity → merchant engagement focus.
- Telangana and Karnataka: very high observed transaction intensity → defend, deepen and monetize.
- Maharashtra: scale + meaningful intensity → defend network, selectively deepen monetization.

WHAT INTERNAL DATA WOULD IMPROVE THIS Replace LTD-base ratios with MAC/MAM, cohort retention, contribution/revenue per active relationship, acquisition cost, merchant category, success rate and district-level competitive density.

11 Merchant Activation Funnel

Public aggregate data cannot answer funnel questions, so the project includes a synthetic analytics layer.



MODELED: illustrative funnel only. No PhonePe internal conversion rates are claimed.

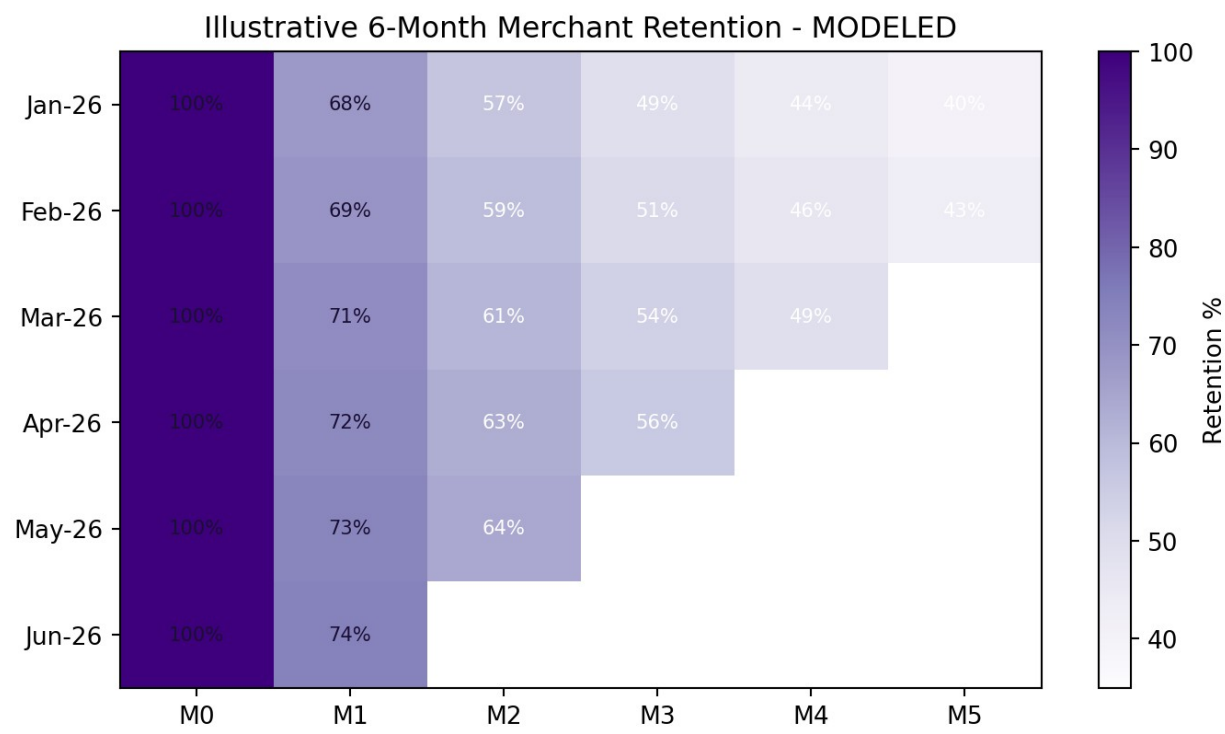
The analytical objective is to locate the first stage where meaningful network value fails to materialize. A merchant QR on a wall is not equivalent to an active merchant; a first transaction is not equivalent to retention.

Diagnostic cuts

- acquisition channel × business category
- state × city tier
- time-to-first-transaction
- first-7-day transaction count
- D30/D90 retention
- device deployment × matched merchant quality
- support/fraud events around activation

12 Cohort Retention

Aggregate MAM can grow while newer cohorts quietly weaken.



MODELED: synthetic cohort matrix for demonstrating SQL and retention analysis.

The repository includes SQL that assigns merchants to onboarding cohorts and calculates M0-M5 retained transacting merchants. In a real PhonePe environment, this should be extended to M12, TPAM retention, TPV retention and revenue retention.

INTERVIEW POINT Cohort analysis controls for calendar-time mix better than a single aggregate retention number. If April acquisition doubled, MAM may rise even while the retention quality of April merchants is poor.

13 Workstream A - Merchant Launchpad

Optimize for retained transacting merchants, not distributed QR codes.

HYPOTHESIS Structured first-30-day onboarding can reduce the QR→first-payment and early-retention drop without relying on blanket transaction subsidies.

Stage	Intervention	Metric
Day 0-1	QR readiness + first-payment test	First successful payment
Day 2-7	Settlement/reconciliation + local-language guidance	7D active rate
Day 8-30	Business insights + reliability nudges	30D retained rate
Post-fit	Device / Udyam / lending or other service eligibility	Retained TPAM + monetization

Experiment

- Control: existing onboarding.
- Treatment: Launchpad sequence with personalized activation tasks.
- Primary: 30-day Retained Merchant Rate.
- Secondary: first payment in 72h, D30 TPAM, active days, support contacts.
- Kill criteria: fraud increase, cost/retained merchant above hurdle, or support burden materially worsens.

14 Workstream B - Consumer Habit Ladder

Move from “more notifications” to next-best-use-case sequencing.

Segment	Illustrative frequency	Product job
Occasional	<10 tx/month	Find one recurring high-fit use case
Developing	10-29 tx/month	Build repeat weekly/monthly loop
Habitual	30+ tx/month	Protect reliability + relevant monetization

The >30 transaction threshold is aligned to PhonePe's disclosed definition of habitual users; <10 and 10-29 are analytical segments introduced for this case.

Example next-use-case logic

- P2P-heavy → merchant scan/pay or recurring utility
- utility-heavy → AutoPay / recurring bill reliability
- merchant-pay-heavy → transit / recurring merchants / credit-on-UPI where eligible
- habitual multi-use-case → security, reliability and relevant financial-service discovery

PRIMARY METRIC Incremental share of MAC crossing the habitual threshold, with transaction success, fraud/disputes, notification opt-out and 60/90-day persistence as guardrails.

15 Workstream C - Merchant Depth & Monetization

Monetize after usage fit, not by indiscriminate cross-sell.

9.19M	₹1,792.8 Cr	₹557.6 Cr	₹452.6 Cr
payment devices Sep-25	FY25 merchant-payments revenue	FY25 lending & insurance revenue	H1FY26 lending & insurance revenue

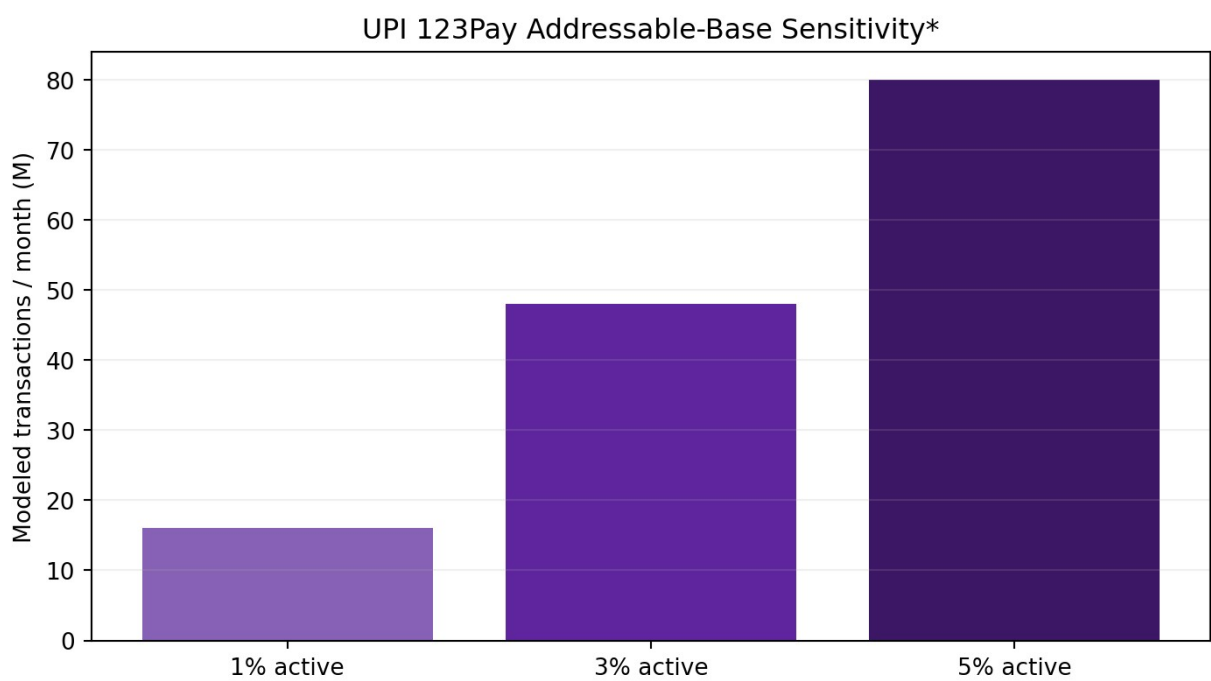
The merchant relationship can become a distribution platform for devices, payment gateway, business tools, formalization support and partner-led lending/insurance. Eligibility should depend on transaction behavior, not just merchant registration.

Level	Merchant state	Potential next layer
1	QR active	First transactions + reliability
2	Retained merchant	Stable TPAM / active days
3	High-frequency	Device / reconciliation tools
4	Business depth	Udyam / payment gateway / settlement services
5	Financial eligibility	Partner lending / insurance where suitable

16 Workstream D - Bharat Access / UPI 123Pay

The next growth frontier should be measured as active digital inclusion.

200M+	No internet	12	31 Aug 2026
feature-phone users cited by PhonePe	SMS-based UPI architecture	Indian-language voice support + English	launch date



MODELED: assumes 8 transactions/month per activated user. Addressable base is a reported PhonePe figure; adoption and usage assumptions are scenarios.

The correct funnel is OEM/pre-bundle → registration → bank linking/PIN setup → first successful payment → D30 retained payer → recurring merchant/P2P use. A pre-installed app without activation is not payments growth.

17 Experimentation Operating System

Each growth idea needs an explicit causal test, guardrails and kill rule.

Test	Population	Primary	Guardrails
Merchant Launchpad	New merchants	30D retained merchant	fraud; cost/retained; support
First-payment nudge	QR ready / zero transaction	7D first payment	opt-out; support
Device eligibility	High-TPAM non-device	incremental TPAM + revenue	device inactivity; churn
Habit Ladder	Developing customers	habitual conversion	success; spam; disputes
123Pay onboarding	Feature-phone users	30D active payer	setup failure; support

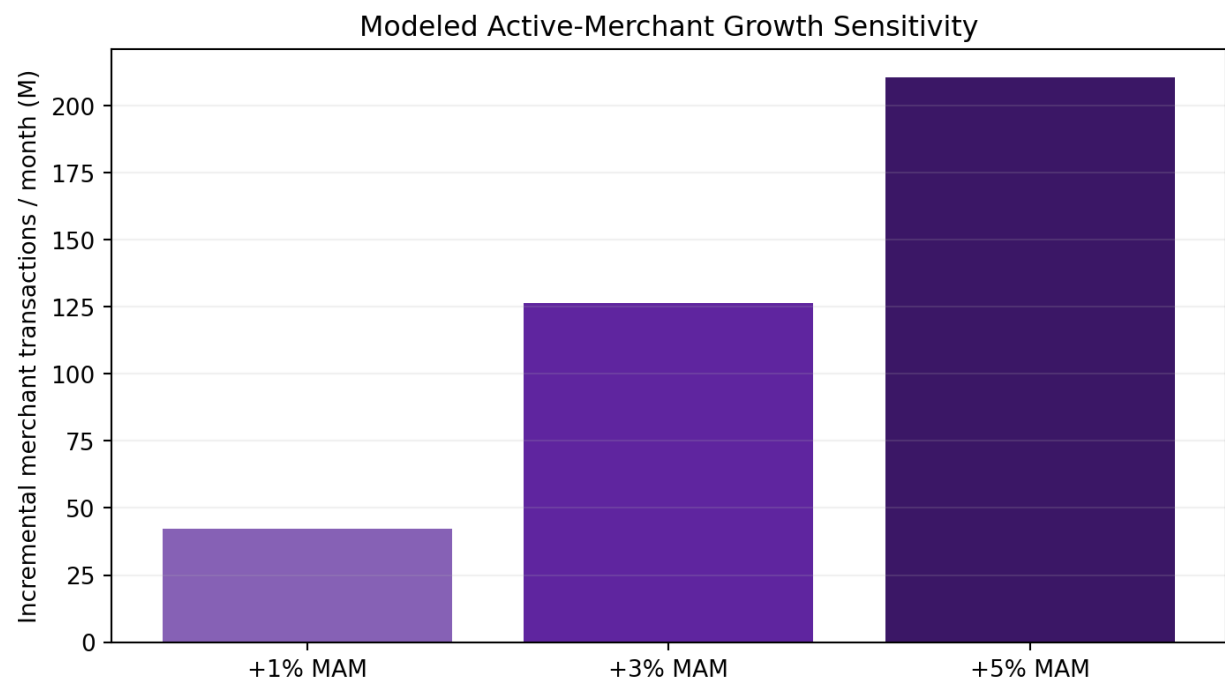
STATISTICAL DISCIPLINE Pre-register the primary metric and minimum detectable effect, randomize at the correct unit, calculate confidence intervals, check sample-ratio mismatch, run long enough to capture weekly/monthly behavior, and evaluate heterogeneous effects before rollout.

Modeled A/B demonstration

The included SQLite dataset contains a synthetic Launchpad experiment with Control vs Treatment outcomes. SQL 06_experiment_readout.sql calculates D30 retention, support-contact and fraud rates. The result is demonstrative, not a PhonePe result.

18 Quantified Opportunity Envelope

Public baselines + explicit scenario assumptions.



MODELED using Sep-2025 MAM = 11.11M and TPAM = 379.16.

Scenario	Incremental MAM	Txn/month	Annualized txn
+1% MAM	0.111M	42.1M	0.51B
+3% MAM	0.333M	126.4M	1.52B
+5% MAM	0.556M	210.6M	2.53B

INTERPRETATION A 3% increase in active merchants at mature-base TPAM corresponds to ~126M additional merchant transactions/month. New merchants will typically ramp below mature TPAM, so this is an opportunity-envelope sensitivity, not a forecast.

Financial-services sensitivity: annualizing H1FY26 Lending & Insurance Distribution revenue gives ~₹905 Cr. A 5% / 10% / 15% uplift on that run rate equals ~₹45 Cr / ₹91 Cr / ₹136 Cr. This is not profit and not company guidance.

19 SQL & Analytics Architecture

The project demonstrates how the strategic thesis becomes an analytical workflow.

REAL PUBLIC LAYER PhonePe Pulse Q2 2026 state aggregates, category mix, company engagement/revenue disclosures.

SYNTHETIC ANALYTICS LAYER 20,000 merchants + monthly activity + modeled experiment outcomes generated only for funnel/cohort/SQL demonstration. Every relevant file is labeled synthetic.

Repository SQL

- 01_state_opportunity.sql - rank geographies using public aggregates
- 02_merchant_activation_funnel.sql - stage conversion by channel
- 03_cohort_retention.sql - M0-M5 active cohort retention
- 04_habitual_segmentation.sql - transaction/TPV depth
- 05_device_conversion.sql - association analysis + causality caveat
- 06_experiment_readout.sql - control/treatment retention + guardrails

What I would request internally

- user/merchant event-level payment logs and success/failure reasons
- merchant acquisition source and acquisition cost
- merchant category, TPV, fee/device revenue and service cost
- device assignment logic and hardware economics
- fraud/dispute/support events
- district/pincode MAC/MAM and cohort retention
- cross-product eligibility and conversion

20 Prioritization & 12-Month Roadmap

Sequence diagnosis → causal tests → selective scale.

Priority	Workstream	Impact	Confidence	Effort
P0	Merchant Launchpad	High	High	Medium
P0	Geographic activation	High	Med-high	Medium
P1	Habit Ladder	High	Medium	Medium
P1	Merchant services eligibility	High	Medium	Med-high
P1 staged	UPI 123Pay	Strategic	Medium	High
Avoid	Blanket registration incentives	Low quality	Low	High

Window	Execution
0-3m	Define activation; build cohorts; state/district diagnostics
3-6m	Launchpad + Habit Ladder + device experiments
6-9m	Scale winners by geography; automate experiment readouts
9-12m	123Pay activation optimization; deeper merchant monetization

RESOURCE PRINCIPLE Spend growth resources where incremental retained activity is strongest, not where registrations are easiest to buy.

21 Risks, Counterarguments & What Would Change My Mind

A strong case study should be falsifiable.

Claim risk	Why incomplete	Correction
Registered/MAC ratio looks low	LTD denominator is not a cohort	Use cohort activation/retention
Device merchants retain better	Selection bias	Randomized eligibility / matching
More transactions = growth	May add fraud/support/cost	Use trust/economic guardrails
Pulse = UPI market share	Metric definitions differ	Use NPCI TPAP data for share
123Pay addressable base is huge	Addressable $\neq$ activated	Measure D30 retained payer
Geo score says UP is best	Weights are modeled	Replace with internal contribution/CAC/MAM

What would change the recommendation?

- If internal data shows merchant activation already saturates quickly, shift P0 toward monetization/service economics.
- If incremental TPAM has negative processing economics, optimize revenue-generating use cases rather than transaction count.
- If device deployment produces weak incremental retention after causal adjustment, reduce hardware expansion.
- If feature-phone setup/support cost overwhelms retained usage, narrow 123Pay rollout to high-readiness segments.

22 Final Recommendation

Acquire reach once; compound value through retained activity, depth, trust and relevant monetization.

P0	P0	P1	P1
Merchant activation	Geo targeting	Habit depth	Merchant monetization

PhonePe should manage the next phase of payments growth as a quality-of-engagement problem. The public data shows scale, but it also shows that the active base is much smaller than lifetime registrations and that active users/merchants are becoming increasingly habitual. This is the exact environment where activation quality and monetization fit matter more than a generic acquisition push.

Resume-ready bullets

- Analyzed PhonePe's 711.7M Q2'26 registered users, 50.7M merchants, payment mix and geographic Pulse data alongside UDRHP engagement metrics to separate acquisition scale from active payment depth.
- Built a state-level activation framework across transaction intensity, merchant density and user scale, identifying high-distribution / lower-engagement markets for targeted activation.
- Designed merchant funnel, cohort-retention and A/B testing frameworks across first transaction → D7 activation → D30 retention → high-frequency usage → device/services monetization; implemented reproducible SQL using clearly labeled synthetic merchant-level data.
- Modeled a 3% lift in active merchants at Sep-2025 TPAM as ~126M incremental merchant transactions/month, with fraud, support, retention and monetization guardrails.

30-second interview pitch

PITCH I treated PhonePe as an activation-and-depth problem rather than a market-share case. It already has 700M+ registered users and 50M+ merchants, but public filings show a much smaller active base while active users and merchants are becoming highly habitual. I combined Pulse geography and transaction data with disclosed engagement metrics to locate activation headroom, then built merchant activation, consumer habit, device/financial-service and feature-phone workstreams. The repository includes SQL funnel/cohort/experiment workflows on synthetic merchant data because public Pulse is aggregated, and every modeled number is separated from reported facts.

23 Methodology & Source Discipline

Why the project is credible in an interview.

- REPORTED: figures quoted directly from PhonePe, NPCI, regulatory filings or current reporting on those filings.
- DERIVED: ratios calculated directly from reported numerators/denominators, e.g. Q2 transactions per LTD registered user.
- MODELED: opportunity scores, funnel conversion, cohort tables, experiment outcomes and impact sensitivities.
- PhonePe Pulse current release restates historical data under new definitions; old and refreshed releases are not joined into one growth series.
- Pulse total transaction definitions are not used as payer-app UPI market-share definitions.
- Synthetic merchant data exists solely because public aggregate data cannot demonstrate merchant-level SQL funnel/cohort analysis.

#	Source	Date	URL
1	PhonePe Updated Draft Red Herring Prospectus-I	2026-01-23	https://www.phonepe.com/apollo/investor-relations/phonepe-limited-updated-draft-red-herring-prospectus-1.pdf
2	PhonePe Pulse	Q2 2026	https://www.phonepe.com/pulse/
3	PhonePe Pulse open dataset	updated through Q2 2026	https://github.com/PhonePe/pulse
4	PhonePe Pulse Metric Definitions	2026	https://www.phonepe.com/pulse/metric-definitions/
5	PhonePe - In the Press / By the Numbers	2026-07	https://www.phonepe.com/press/
6	UPI Product Statistics	2026	https://www.npci.org.in/product/upi/product-statistics
7	India preparing rollout of agentic payments on UPI	2026-09-01	https://www.reuters.com/world/india/india-preparing-rollout-agentic-payments-upi-sources-say-2026-09-01/
8	PhonePe reaches 50M registered merchants	2026-04-28	https://www.phonepe.com/press/phonepe-reaches-50-million-registered-merchants-across-india/
9	PhonePe crosses 700M registered users	2026-04-29	https://www.phonepe.com/press/phonepe-surpasses-700-million-registered-users-accelerates-growth-momentum/
10	PhonePe UPI 123Pay launch	2026-08-31	https://www.phonepe.com/press/phonepe-launches-upi-payments-for-indias-feature-phone-users
11	PhonePe FY26 financials	2026-07-25	https://www.business-standard.com/companies/quarterly-results/phonepe-fy26-net-loss-widens-to-rs-2-792-cr-despite-11-5-pc-revenue-growth-126072501086_1.html
12	PhonePe PulsePro launch	2026-07-30	https://www.phonepe.com/press/phonepe-launches-pulsepro-actionable-data-intelligence-for-indian-businesses/

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